

Zhaoyang Zhang

Curriculum Vitae

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Google Scholar

Education

- 2019.09–2024.01 **Ph.D. in Electronic Engineering**, *Multimedia Lab*, The Chinese University of Hong Kong.
- Under supervision of **Prof. Xiaogang Wang** and **Prof. Ping Luo**
- 2014.06–2018.06 **B.Sc. in Computer Science**, Wuhan University.
- Class of Outstanding Engineers (Ranking 2/294), under supervision of **Prof. Xiaohui Cui**

Research Interests

- **Efficient Reinforcement Learning**
e.g., RL-aware distillation and RL for image editing
- **Image/Video Generation and Editing**
e.g., generative image and video inpainting and editing
- **Efficient Deep Learning**
e.g., network quantization and pruning
- **Sample-based Dynamic Learning**
e.g., dynamic architectures and neural architecture search
- **Computational Photography**
e.g., image denoising and restoration

Working Experiences

- 2025.05–Present **Applied Scientist II**, Fundamental Research Team, AWS AI Lab, Bellevue, WA.
- **Role:** Working on **RL** and **post-training** for agentic AI.
 - **RL-aware Distillation for LLM Reasoning.**
 - Role: First author and major contributor.
 - Submitted to ICML 2026.
 - **Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes.**
 - Role: Corresponding author.
 - Project page: *Talk2Move*
 - **Collaborators:** **Prof. Zhuowen Tu** and **Prof. Stefano Soatto**
- 2024.01–2025.03 **Senior Research Scientist, Tech Lead**, ARC Lab, Tencent.
- **Role:** Working on image and video generation and editing.
 - **BlobCtrl: A Unified and Flexible Framework for Element-level Image Generation and Editing.**
 - SIGGRAPH Asia 2025, Project Lead.
 - Project page: *BlobCtrl*

- **VideoPainter: Any-length Video Inpainting and Editing with Plug-and-Play Context Control.**
 - SIGGRAPH 2025, Project Lead.
 - Project page: *VideoPainter*
 - **FlexiAct: Towards Flexible Action Control in Heterogeneous Scenarios.**
 - SIGGRAPH 2025, Project Lead.
 - Project page: *FlexiAct*
 - **Cobra: Efficient Line Art Colorization with Broader References.**
 - SIGGRAPH 2025, Project Lead.
 - Project page: *Cobra*
 - **BrushEdit: All-In-One Image Inpainting and Editing.**
 - TPAMI under review, Project Lead.
 - Project page: *BrushEdit*
- 2022.09–2023.03 **Applied Scientist Intern**, AWS AI Lab, Seattle, WA.
- Musketeer, a generalist vision-language model with task explanation prompts.
 - Under supervision of **Prof. Zhuowen Tu** and **Prof. Stefano Soatto**
- 2020.06–2021.02 **Research Intern**, SenseBrain Ltd, Hong Kong.
- Efficient deep learning for computational photography. ISP optimization, e.g., HDR reconstruction, white balance, denoising, and illumination enhancement.
 - Under supervision of **Prof. Jinwei Gu**
- 2018.09–2019.06 **Research Assistant**, The Chinese University of Hong Kong.
- Neural architecture search.
 - Under supervision of **Prof. Ping Luo**

Publications

Reinforcement Learning.

- **Z Zhang**, S Jiang, Y Shen, Y Zhang, D Ram, S Yang, W Xia, Z Tu, S Soatto, “*RL-aware Distillation for LLM Reasoning.*” , submitted to **ICML 2026**.
- J Tan, **Z Zhang**, Y Shen, J Cai, S Yang, J Wu, W Xia, Z Tu, S Soatto, “*Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes*” , submitted to **CVPR 2026**, Corresponding author.

Image/Video Generation - Main Contributor.

- S Zhang, J Zhuang, **Z Zhang**, Y Shan, Y Tang, “*BlobCtrl: A Unified and Flexible Framework for Element-level Image Generation and Editing*” , accepted at **SIGGRAPH Asia 2025**, Project Lead.
- S Zhang, J Zhuang, **Z Zhang**, Y Shan, Y Tang, “*FlexiAct: Towards Flexible Action Control in Heterogeneous Scenarios*” , accepted at **SIGGRAPH 2025**, Project Lead.
- J Zhuang, L Li, X Ju, **Z Zhang**, C Yuan, Y Shan, “*Cobra: Efficient Line Art Colorization with Broader References*” , accepted at **SIGGRAPH 2025**, Project Lead.
- Y Bian, **Z Zhang**, X Ju, M Cao, L Xie, Y Shan, Q Xu, “*VideoPainter: Any-length Video Inpainting and Editing with Plug-and-Play Context Control*” , accepted at **SIGGRAPH 2025**, Project Lead.

- Y Li, X Wang, **Z Zhang**, Z Wang, Z Yuan, L Xie, Y Shan, Y Zou, “*Image Conductor: Precision Control for Interactive Video Synthesis*” , accepted at **AAAI 2025**, Project Lead.
- Y Bian, **Z Zhang**, X Ju, M Cao, L Xie, Y Shan, Q Xu, “*NVComposer: Boosting Generative Novel View Synthesis with Multiple Sparse and Unposed Images*” , accepted at **CVPR 2025**, Project Lead.

Image/Video Generation - Other Contributions.

- L Li, G Wang, **Z Zhang**, Y Li, X Li, Q Dou, J Gu, T Xue, Y Shan, “*Tooncomposer: Streamlining Cartoon Production with Generative Post-keyframing*” , submitted to **ICLR 2026**.
- Y Li, X Li, **Z Zhang**, Y Bian, G Liu, X Li, J Xu, W Hu, Y Liu, L Li, J Cai, Y Zou, Y He, Y Shan, “*IC-Custom: Diverse Image Customization via In-Context Learning*” , submitted to **ICLR 2026**.
- X Ju*, Y Gao*, **Z Zhang***, Z Yuan, X Wang, A Zeng, Y Xiong, Q Xu, Y Shan, “*MiraData: A Large-Scale Video Dataset with Long Durations and Structured Captions*” , accepted at **NeurIPS 2024**, Project Lead.
- C Mou, M Cao, X Wang, **Z Zhang**, Ying Shan, Jian Zhang, “*ReVideo: Remake a Video with Motion and Content Control*” , accepted at **NeurIPS 2024**.
- Y Jiang*, **Z Zhang***, T Xue, J Gu, “*AutoDIR: Automatic All-in-One Image Restoration with Latent Diffusion*” , accepted at **ECCV 2024**.

Efficient AI - Main Contributor.

- **Z Zhang**, W Shao, Y Ge, Y Shan, J Gu, X Wang, P Luo, “*Cached Transformers: Improving Transformers with Differentiable Memory Cache*” , accepted at **AAAI 2024**.
- **Z Zhang**, Y Jiang, W Shao, X Wang, P Luo, K Lin, J Gu, “*Real-time Controllable Denoising for Image and Video*” , accepted at **CVPR 2023**.
- **Z Zhang**, Y Jiang, J Jiang, X Wang, P Luo, J Gu, “*STAR: A Semantic-aware Transformer for Real-time Image Enhancement*” , accepted at **ICCV 2021**.
- **Z Zhang**, W Shao, X Wang, P Luo, “*Differentiable Dynamic Quantization with Mixed Precision and Adaptive Resolution*” , accepted at **ICML 2021 (Spotlight)**.
- **Z Zhang**, J Li, W Shao, Z Peng, R Zhang, X Wang, P Luo, “*Differentiable Learning-to-group Channels via Groupable Convolutional Neural Networks*” , accepted at **ICCV 2019**.
- **Z Zhang**, Z Kuang, P Luo, L Feng, W Zhang, “*Temporal Sequence Distillation: Towards Few-Frame Action Recognition in Videos*” , accepted at **ACM MM 2018**.

Efficient AI - Other Contributions.

- W Shao*, M Chen*, **Z Zhang**, P Xu, L Zhao, Z Li, K Zhang, P Gao, Y Qiao, P Luo, “*OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models*” , accepted at **ICLR 2024 (Spotlight)**.
- J Li, W Liu, **Z Zhang**, J Wang, T Lee, “*Model Compression for DNN-based Speaker Verification Using Weight Quantization*” , accepted at **INTERSPEECH 2023**.
- W Shao, X Zhao, Y Ge, **Z Zhang**, L Yang, X Wang, Y Shan, P Luo, “*Not All Models Are Equal: Predicting Model Transferability in a Self-challenging Fisher Space*” , accepted at **ECCV 2022**.

- W Shao, Y Ge, **Z Zhang**, X Xu, X Wang, Y Shan, P Luo, “*Dynamic Token Normalization Improves Vision Transformer*” , accepted at **ICLR 2022**.
- C Tao, R Lin, Q Chen, **Z Zhang**, ..., P Luo, “*FAT: Learning Low-Bitwidth Parametric Representation via Frequency-Aware Transformation*” , accepted at **IEEE-TNNLS**.
- X Yue, Z Kuang, **Z Zhang**, Z Chen, P He, Y Qiao, W Zhang, “*Boosting up Scene Text Detectors with Guided CNN*” , accepted at **BMVC 2018 (Oral)**.
- W Shao, H Yu, **Z Zhang**, H Xu, Z Li, P Luo, “*BWCP: Probabilistic Learning-to-Prune Channels for ConvNets via Batch Whitening*”.
- W Li, **Z Zhang**, X Wang, P Luo, “*Adax: Adaptive Gradient Descent with Exponential Long Term Memory*”.

Honors & Awards

- **1st place winner** (Project Lead), GenAI Media Generation Challenge Workshop @ CVPR 2024. Tech report: <https://arxiv.org/pdf/2407.1139>
- National scholarship, Wuhan University, 2014-2016
- Outstanding intern, SenseBrain, 2019-2020
- Outstanding graduates, Wuhan University, 2018

Teaching Experiences

- 2020-2021 **Teaching Assistant**, ENGG 2740A: Differential Equations, CUHK.
- 2020-2021 **Teaching Assistant**, ENGG 2720: Complex Analysis, CUHK.
- 2019-2020 **Teaching Assistant**, ENGG 2420A: Complex Analysis and Differential Equations, CUHK.